

## SAFETY DATA SHEET

Creation Date 29-September-2006

Revision Date 27-March-2024

Revision Number 4

### 1. Identification

**Product Name** Lawesson's Reagent

**Cat No. :** A14530

**CAS-No** 19172-47-5  
**Synonyms** 2,4-Bis(4-methoxyphenyl)-1,3-dithia-2,4-diphosphetane-2,4-disulfide

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
 112 Colonnade Road,  
 Ottawa, ON K2E 7L6,  
 Canada  
 Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                               |            |                            |
|-------------------------------------------------------------------------------|------------|----------------------------|
| <b>Substances/mixtures which, in contact with water, emit flammable gases</b> | Category 2 |                            |
| <b>Acute Inhalation Toxicity</b>                                              | Category 1 | (based on evolved PH3 gas) |
| <b>Health Hazards Not Otherwise Classified</b>                                | Category 1 |                            |
| In contact with water, releases gases which are toxic if inhaled              |            |                            |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

In contact with water releases flammable gas

Fatal if inhaled

In contact with water, releases gases which are toxic if inhaled



### Precautionary Statements

#### Prevention

Do not allow contact with water  
Do not breathe dust/fumes/gas/mist/vapours/spray  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Handle and store contents under inert gas. Protect from moisture

#### Response

IF INHALED: Remove person to fresh air and keep comfortable for breathing  
Call a POISON CENTER/ doctor  
IF ON SKIN: Brush off loose particles from skin. Immerse in cool water  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

#### Storage

Store in a dry place. Store in a closed container  
Store in a well-ventilated place. Keep container tightly closed

#### Disposal

Dispose of contents/container to an approved waste disposal plant

### Other Hazards

Stench

## 3. Composition/Information on Ingredients

| Component                                                                | CAS-No     | Weight % |
|--------------------------------------------------------------------------|------------|----------|
| 1,3,2,4-Dithiadiphosphetane,<br>2,4-bis(4-methoxyphenyl)-, 2,4-disulfide | 19172-47-5 | <=100    |

## 4. First-aid measures

|                                                               |                                                                                                                   |
|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                                         | If symptoms persist, call a physician.                                                                            |
| <b>Eye Contact</b>                                            | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.   |
| <b>Skin Contact</b>                                           | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician. |
| <b>Inhalation</b>                                             | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.      |
| <b>Ingestion</b>                                              | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.             |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | None reasonably foreseeable.<br>Treat symptomatically                                                             |

## 5. Fire-fighting measures

|                                         |                                                                                                                                                                           |
|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Use extinguishing measures that are appropriate to local circumstances and the surrounding environment. CO <sub>2</sub> , dry chemical, dry sand, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                                                                                                                  |
| <b>Flash Point</b>                      | No information available                                                                                                                                                  |
| <b>Method -</b>                         | No information available                                                                                                                                                  |
| <b>Autoignition Temperature</b>         | Not applicable                                                                                                                                                            |
| <b>Explosion Limits</b>                 |                                                                                                                                                                           |
| <b>Upper</b>                            | No data available                                                                                                                                                         |
| <b>Lower</b>                            | No data available                                                                                                                                                         |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                                                                                                                  |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                                                                                                                  |

**Specific Hazards Arising from the Chemical**

Water reactive. Contact with water liberates toxic gas. Produce flammable gases on contact with water. Contact with water liberates toxic gas.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Sulfur oxides. Phosphorus trihydride (phosphine).

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
2

**Flammability**  
2

**Instability**  
1

**Physical hazards**  
W

## 6. Accidental release measures

|                                             |                                                                                                                                            |
|---------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                          |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                               |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal. Do not expose spill to water. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                 |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. Do not allow contact with water.                            |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Keep from any possible contact with water. Keep away from water or moist air. Store under an inert atmosphere. Incompatible Materials. Water. Strong oxidizing agents. |

## 8. Exposure controls / personal protection

|                             |                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>  | This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.                                                                                                                                                                                                                       |
| <b>Engineering Measures</b> | Ensure adequate ventilation, especially in confined areas.<br>Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |

**Personal protective equipment**

**Eye Protection**  
**Hand Protection**

Goggles  
 Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                               |                                  |
|-----------------------------------------------|----------------------------------|
| <b>Physical State</b>                         | Powder Solid                     |
| <b>Appearance</b>                             | Light yellow                     |
| <b>Odor</b>                                   | Stench                           |
| <b>Odor Threshold</b>                         | No information available         |
| <b>pH</b>                                     | Not applicable                   |
| <b>Melting Point/Range</b>                    | 228 - 231 °C / 442.4 - 447.8 °F  |
| <b>Boiling Point/Range</b>                    | No information available         |
| <b>Flash Point</b>                            | No information available         |
| <b>Evaporation Rate</b>                       | Not applicable                   |
| <b>Flammability (solid,gas)</b>               | No information available         |
| <b>Flammability or explosive limits</b>       |                                  |
| Upper                                         | No data available                |
| Lower                                         | No data available                |
| <b>Vapor Pressure</b>                         | No information available         |
| <b>Vapor Density</b>                          | Not applicable                   |
| <b>Specific Gravity</b>                       | No information available         |
| <b>Solubility</b>                             | Decomposes in contact with water |
| <b>Partition coefficient; n-octanol/water</b> | No data available                |
| <b>Autoignition Temperature</b>               | Not applicable                   |
| <b>Decomposition Temperature</b>              | No information available         |
| <b>Viscosity</b>                              | Not applicable                   |
| <b>Molecular Formula</b>                      | C14 H14 O2 P2 S4                 |
| <b>Molecular Weight</b>                       | 404.45                           |

## 10. Stability and reactivity

|                                         |                                                                                                           |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | Yes                                                                                                       |
| <b>Stability</b>                        | Moisture sensitive.                                                                                       |
| <b>Conditions to Avoid</b>              | Incompatible products. Exposure to moist air or water. Exposure to moisture.                              |
| <b>Incompatible Materials</b>           | Water, Strong oxidizing agents                                                                            |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Sulfur oxides, Phosphorus trihydride (phosphine) |
| <b>Hazardous Polymerization</b>         | No information available.                                                                                 |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                                             |

## 11. Toxicological information

### Acute Toxicity

**Product Information** No acute toxicity information is available for this product

**Component Information**

**Toxicologically Synergistic Products** No information available

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component                                                                | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------------------------------------------------|------------|------------|------------|------------|------------|------------|
| 1,3,2,4-Dithiadiphosphetane,<br>2,4-bis(4-methoxyphenyl)-, 2,4-disulfide | 19172-47-5 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Do not empty into drains. Reacts with water so no ecotoxicity data for the substance is available.

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment.

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

**UN-No** UN3134  
**Proper Shipping Name** Water-reactive solid, toxic, n.o.s.  
**Hazard Class** 4.3  
**Subsidiary Hazard Class** 6.1  
**Packing Group** II

#### TDG

**UN-No** UN3134  
**Proper Shipping Name** Water-reactive solid, toxic, n.o.s.  
**Hazard Class** 4.3  
**Subsidiary Hazard Class** 6.1  
**Packing Group** II

#### IATA

**UN-No** UN3134  
**Proper Shipping Name** WATER-REACTIVE SOLID, TOXIC, N.O.S.\*  
**Hazard Class** 4.3  
**Subsidiary Hazard Class** 6.1  
**Packing Group** II

#### IMDG/IMO

**UN-No** UN3134  
**Proper Shipping Name** Water-reactive solid, toxic, n.o.s.  
**Hazard Class** 4.3  
**Subsidiary Hazard Class** 6.1  
**Packing Group** II

### 15. Regulatory information

#### International Inventories

| Component                                                             | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|-----------------------------------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 1,3,2,4-Dithiadiphosphetane, 2,4-bis(4-methoxyphenyl)-, 2,4-disulfide | 19172-47-5 | -   | X    | X    | ACTIVE                                        | 242-855-4 | -      | -   |

| Component                                                             | CAS-No     | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------------------------------------------------------------|------------|-------|------|------|------|------|------|-------|-------|
| 1,3,2,4-Dithiadiphosphetane, 2,4-bis(4-methoxyphenyl)-, 2,4-disulfide | 19172-47-5 | X     | -    | -    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances  
KECL - Korean Existing and Evaluated Chemical Substances  
ENCS - Japanese Existing and New Chemical Substances  
AICS - Australian Inventory of Chemical Substances  
PICCS - Philippines Inventory of Chemicals and Chemical Substances

**Canada**

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

**Other International Regulations**

Authorisation/Restrictions according to EU REACH Not applicable

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component                                                             | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------------------------------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 1,3,2,4-Dithiadiphosphetane, 2,4-bis(4-methoxyphenyl)-, 2,4-disulfide | 19172-47-5 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                                                             | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 1,3,2,4-Dithiadiphosphetane, 2,4-bis(4-methoxyphenyl)-, 2,4-disulfide | 19172-47-5 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

**16. Other information**

**Prepared By** Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

**Creation Date** 29-September-2006  
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**Print Date** 27-March-2024  
**Revision Summary** New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**